NLP PRACTICAL CEC

Intelligent Legal Document Analysis and Risk Detection

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Problem Statement and Objectives

Problem

Legal contracts are long, dense and full of jargon. Reviewing them by hand is slow, and risky or missing clauses are easy to overlook.

Objectives

- Ingest TXT, PDF and DOCX contracts
- Segment and classify clauses into 14 categories
- Extract legal entities
- Score risk and flag missing clauses
- Summarize, simplify and answer questions

Introduction and Motivation

Hard to read

Contracts hide obligations in long, technical sentences.

Costly to review

Legal review takes time and expertise many people lack.

Risk goes unseen

Liability, termination and indemnity terms need quick attention.

Motivation: give anyone a fast, plain-English first pass over a contract using classical NLP.

Dataset Description and Source

Data	Description	Source
Test contract	Sample legal agreement used for testing	samples/sample_contract.txt
User input	Pasted text or uploaded .txt, .pdf, .docx files	Read with pypdf and python-docx
Clause labels	14 legal clause categories for the classifier	[Add training data source]
Entities	Parties, dates, amounts, jurisdictions, laws, durations, percentages	Extracted from contract text

NLP Techniques and Algorithms

Segmentation
Numbered headings, Roman numerals and sentence groups.

NER
Parties, dates, money, jurisdictions, laws, durations, percentages.

Classification
TF-IDF with calibrated Logistic Regression and keyword boosting.

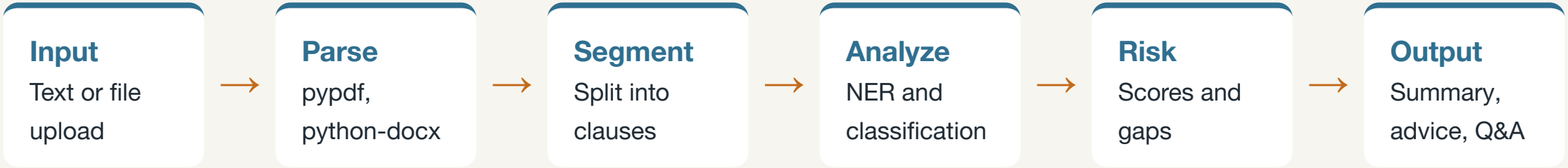
Risk Scoring
Severity on a 0–10 scale, missing clauses and risky phrasing.

Summarization
Extractive, TF-IDF centroid-based.

Simplification
Plain-English rewrite with negotiation tips.

Semantic Search
TF-IDF cosine similarity, optional SBERT.

System Architecture and Workflow



The browser sends a request to **POST /api/analyze**. The Flask app calls the NLP engine and returns JSON that the web interface displays.

Implementation and Tools Used

Backend

Python and Flask routes with a single analysis API.

NLP Engine

Scikit-Learn for TF-IDF, Logistic Regression and cosine similarity.

Parsing

pypdf for PDF and python-docx for Word files.

Frontend

HTML, CSS and JavaScript with AJAX calls.

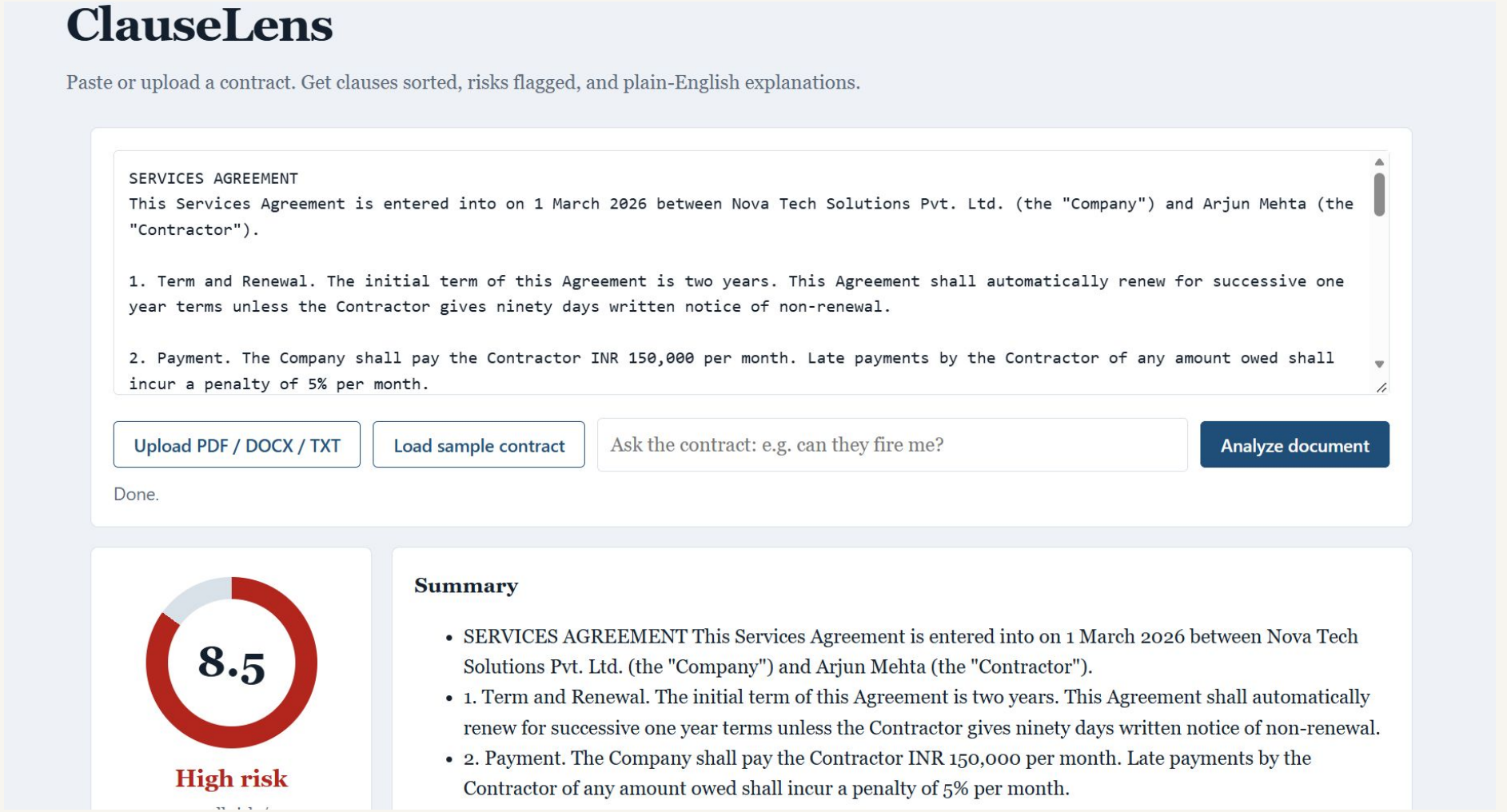
Optional

Sentence Transformers and spaCy for richer embeddings.

Run

pip install -r requirements.txt, then python app.py.

Results, Evaluation and Screenshots



Results, Evaluation and Screenshots

years after termination and shall not solicit the Company's customers.

328 words | 13 clauses | ~1 min read | 5 high | 5 medium | 3 low

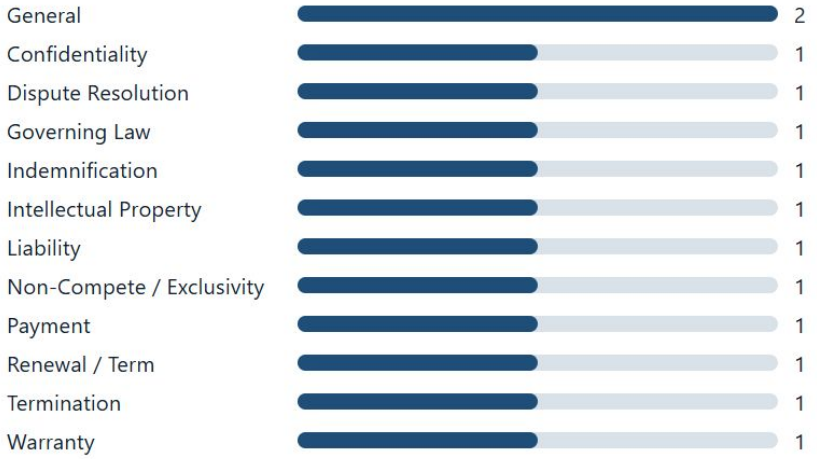
Top risks found

- One-sided assignment × 2
- Broad non-compete / exclusivity × 2
- Automatic renewal / lock-in × 1
- Heavy penalties / late fees × 1
- Unilateral termination without cause × 1
- Vague / subjective wording × 1

Possibly missing clauses

None of the standard clauses are missing.

Clause mix



Key entities

INR 150,000 MONEY | INR 10,000 MONEY | 1 March 2026 DATE | two years DURATION | one year DURATION | ninety days DURATION | five years DURATION | three years DURATION | 5% PERCENT | Nova Tech Solutions Pvt. Ltd ORG | Payment. The Company ORG | Termination. The Company ORG | The Company ORG | Warranty. The Company ORG | Modification. The Company ORG | laws of India LAW/JURISDICTION | courts of Mumbai LAW/JURISDICTION | Company DEFINED TERM | Contractor DEFINED TERM

Results, Evaluation and Screenshots

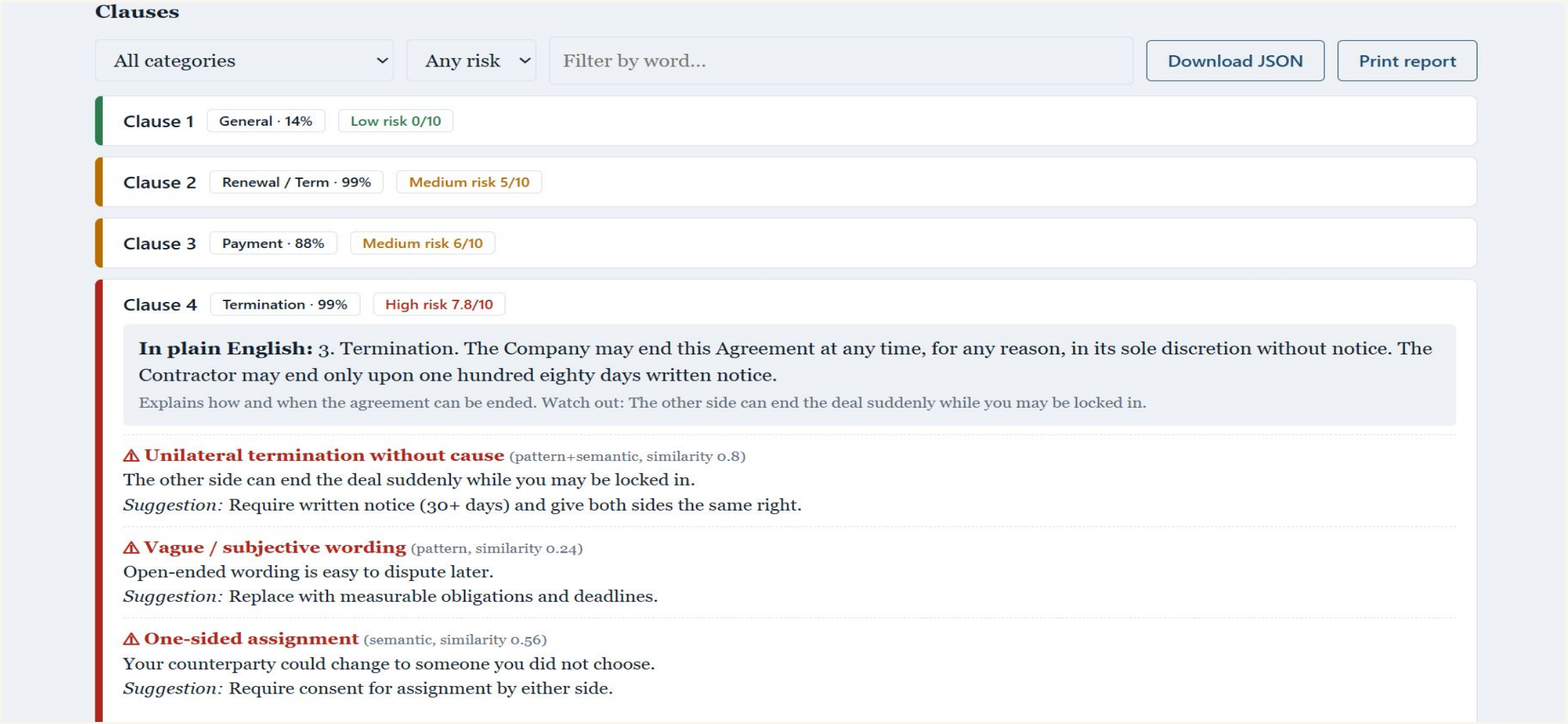
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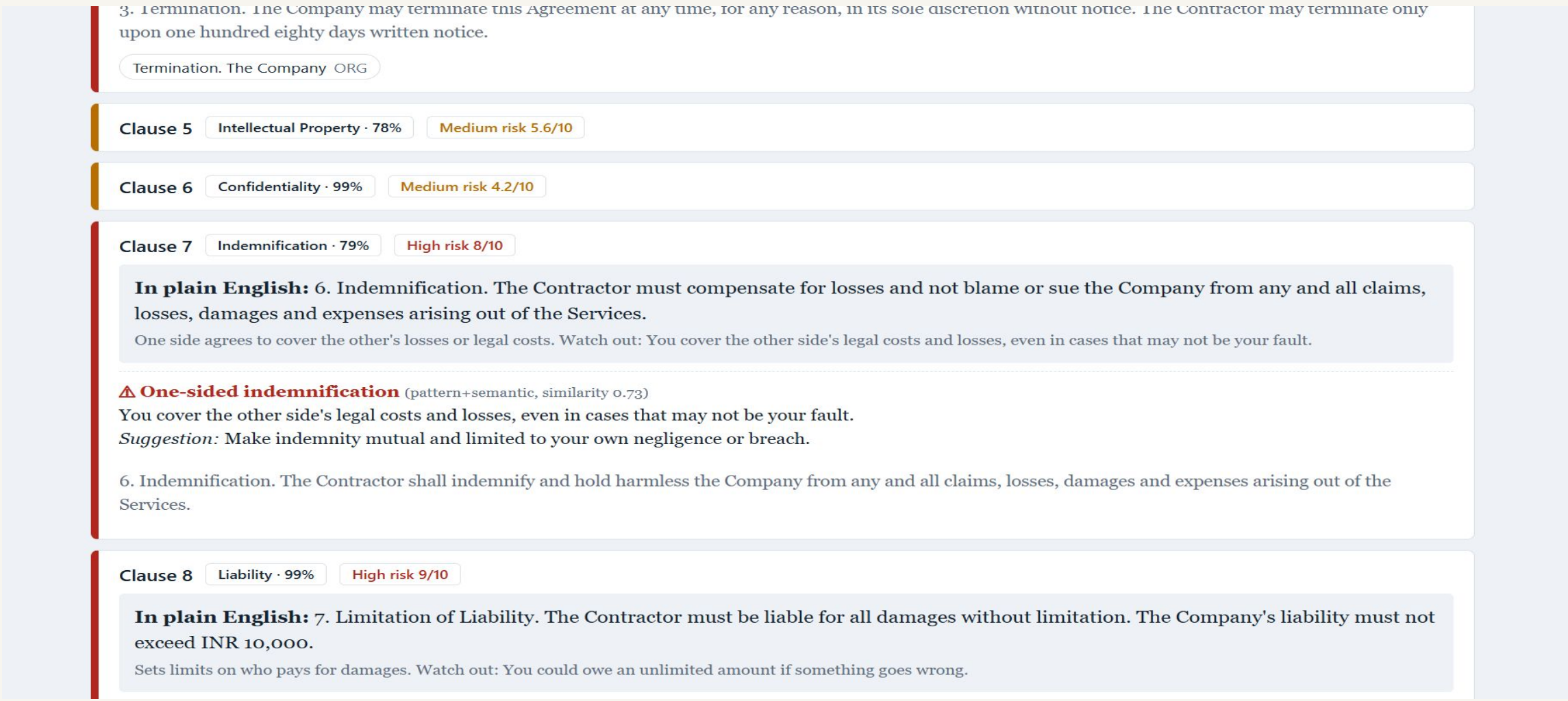
Obligations

- **This Agreement**: This Agreement shall automatically renew for successive one year terms unless the Contractor gives ninety days written notice of non-renewal.
- **The Company**: The Company shall pay the Contractor INR 150,000 per month.
- **Late payments by the Contractor of any amount owed**: Late payments by the Contractor of any amount owed shall incur a penalty of 5% per month.
- **The Contractor**: The Contractor shall keep all Confidential Information strictly confidential and shall not disclose it to any third party during and for five years after the term.
- **The Contractor**: The Contractor shall indemnify and hold harmless the Company from any and all claims, losses, damages and expenses arising out of the Services.
- **The Contractor**: The Contractor shall be liable for all damages without limitation.
- **The Company's liability**: The Company's liability shall not exceed INR 10,000.
- **The Contractor**: The Contractor shall not engage in any competing business anywhere in India for three years after termination and shall not solicit the Company's customers.
- **All disputes**: All disputes shall be settled by binding arbitration in Mumbai and the Contractor waives the right to a jury trial or class action.
- **This Agreement**: This Agreement shall be governed by the laws of India and the courts of Mumbai shall have exclusive jurisdiction.

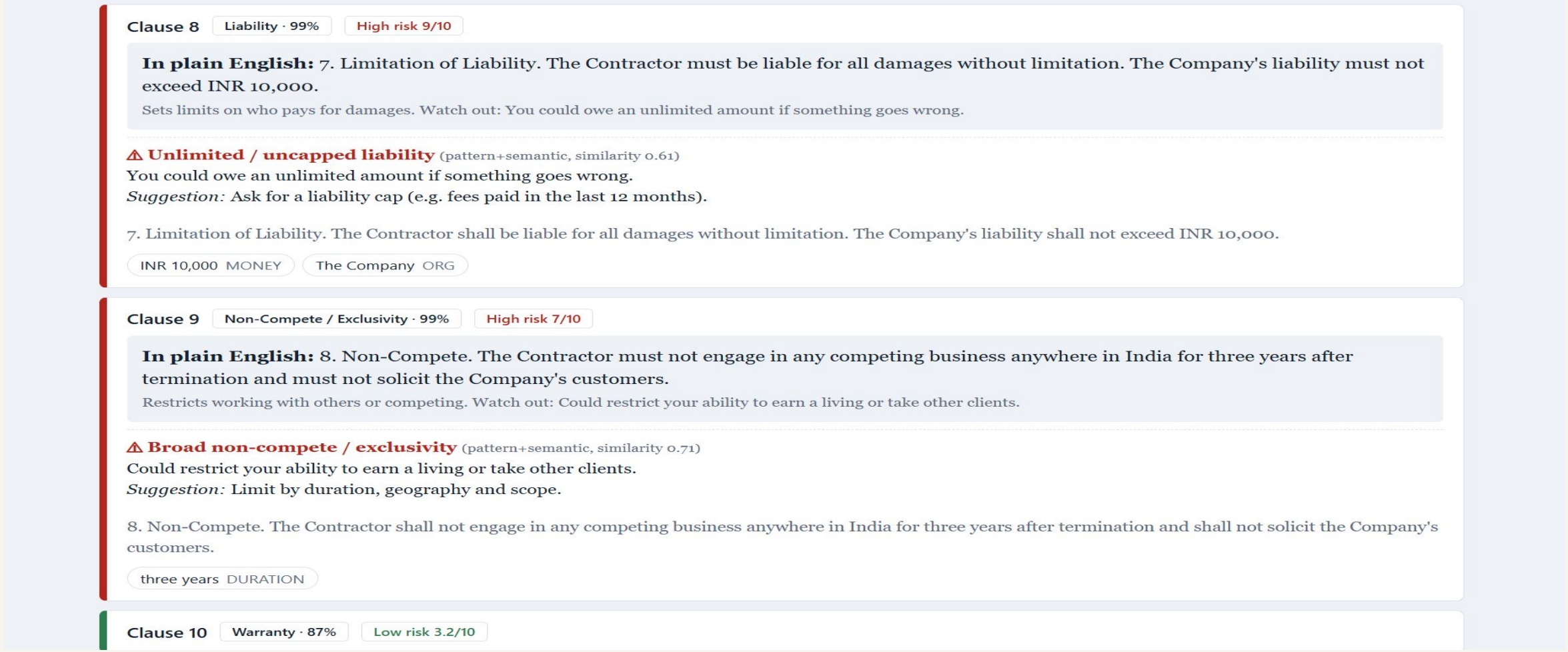
Results, Evaluation and Screenshots



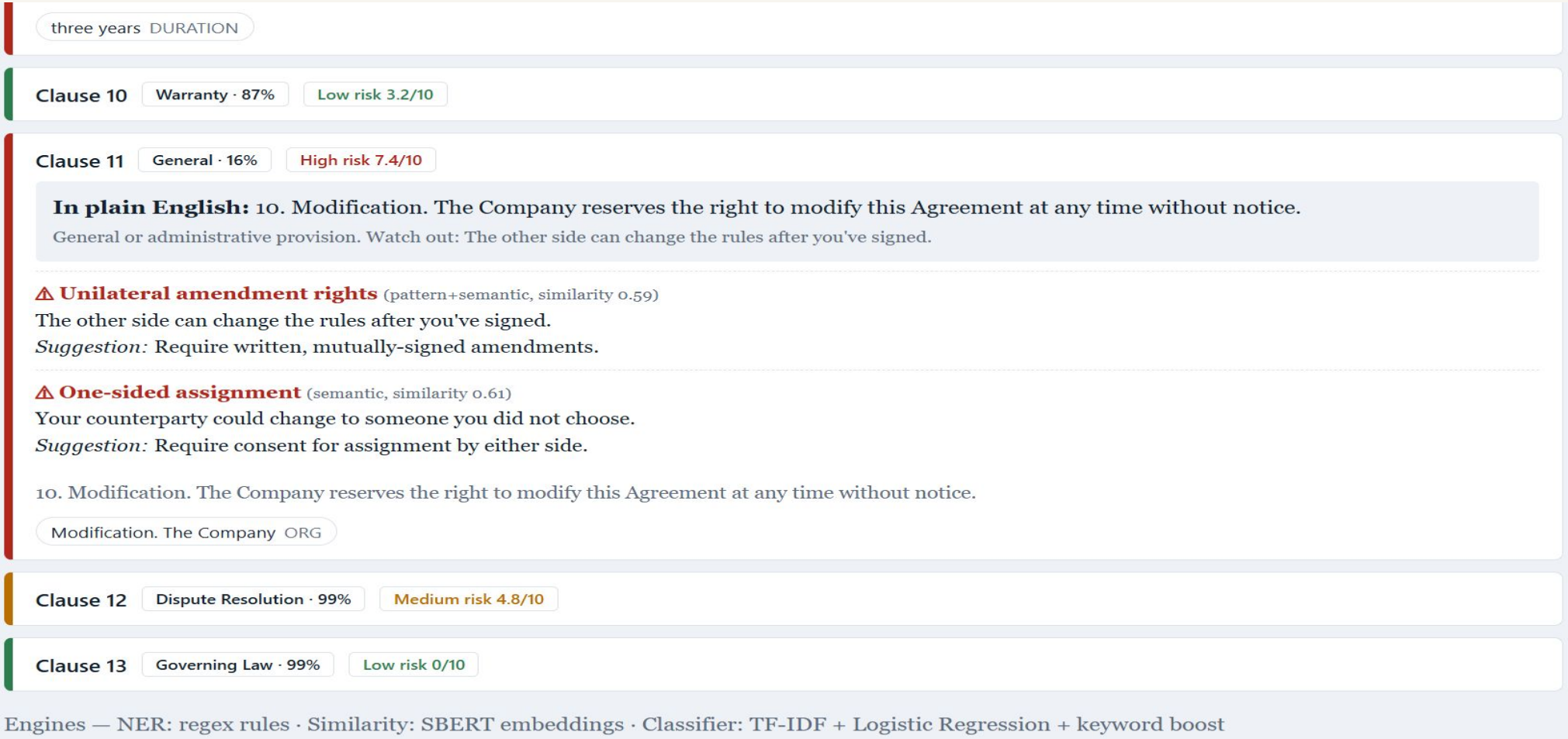
Results, Evaluation and Screenshots



Results, Evaluation and Screenshots



Results, Evaluation and Screenshots



Results, Evaluation and Screenshots

Output produced

Clauses with category, risk score and level, entities and advice, plus overall risk, summary and missing clauses.

Evaluation

Classifier accuracy: [__%]. Tested on: [__ contracts]. Add your measured values.

Conclusion and Future Scope

Conclusion

- Classical NLP gives a fast, explainable first review
- One pipeline covers extraction, risk, summary and Q&A
- Output stays in plain English

Future Scope

- Train on larger labeled contract sets
- Add transformer-based classification and SBERT search
- Support more languages and contract types
- Compare versions of a contract

References

- Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12.
- Reimers and Gurevych (2019). Sentence-BERT. EMNLP-IJCNLP.
- Flask documentation: flask.palletsprojects.com
- pypdf and python-docx documentation
- spaCy documentation: spacy.io
- Project repository: github.com/ravalharshit25-a11y/nlp_practical_cec